

Navigating Stakeholder Tensions in Spatial Computing: A Participatory Design Approach for Inclusive Mixed Reality Environments

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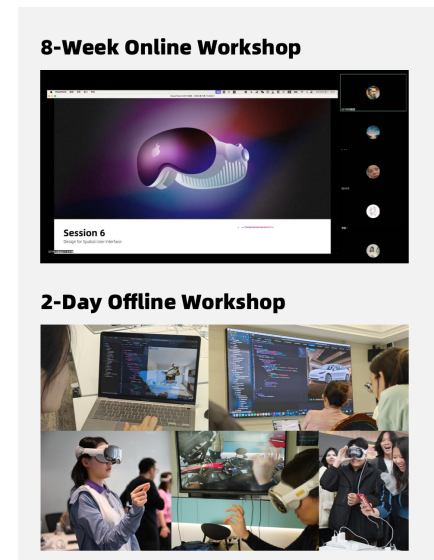
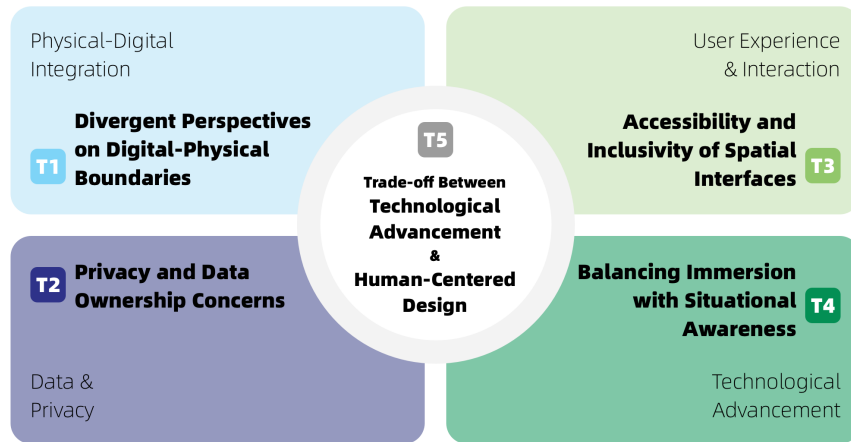


Figure 1: Five Critical Tensions among Participants

Abstract

Spatial computing technologies—encompassing augmented, virtual, and mixed reality—are increasingly adopted across industries. Yet designing these systems often overlooks the complex interplay of stakeholder incentives. This paper presents a participatory design framework integrating technological innovations (e.g., exploring Apple Vision Pro and SwiftUI prototyping) and conceptual contributions (e.g., co-design workshops) to understand and

address stakeholder tensions. We identified five critical tensions through eight online and two in-person participatory sessions with over 80 participants: divergent perspectives on digital-physical boundaries, privacy and data ownership, accessibility, and inclusivity; balancing immersion with situational awareness; and reconciling rapid technological advancement with human-centered design. We discuss how these tensions intersect and offer actionable strategies—contextual customization and layered accessibility—to make spatial computing experiences more inclusive and ethically grounded. Beyond Apple Vision Pro and SwiftUI, we demonstrate how our participatory methodology can be adapted to other platforms, ensuring broader applicability to future spatial computing endeavors.

CCS Concepts

• Human-centered computing → Mixed / augmented reality.

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1 Introduction

Spatial computing, encompassing virtual reality (VR), augmented reality (AR), and mixed reality (MR), redefines how users interact with digital content in physical environments. Applications range from collaborative industrial simulations to immersive educational platforms. However, current research often narrows on usability features or hardware capabilities without fully accounting for the stakeholder tensions in real-world deployments. Designers, developers, institutional staff, and end users bring unique incentives—such as innovation goals, budget constraints, regulatory compliance, or comfort and accessibility requirements—leading to conflicting priorities that impede adoption or result in inequitable user experiences.

While prior studies have explored usability (e.g., interface metaphors, motion sickness), data privacy (e.g., ethical data capture), and accessibility (e.g., hardware options for diverse users), few have examined how these issues intersect in a multi-stakeholder context. Ethical and sociotechnical critiques of spatial computing remain relatively fragmented, often limited to single stakeholder groups (e.g., end users or just developers). This gap highlights the need for a holistic approach that systematically addresses tensions across these roles, acknowledging that a single design decision (e.g., how data is processed) can have far-reaching impacts (e.g., on institutional policies and user acceptance) [9, 15].

Against this backdrop, our research investigates the tensions inherent in mixed reality systems by engaging multiple stakeholder groups through participatory methods. This approach captures a breadth of perspectives and facilitates collaborative prototyping, enabling participants to co-create potential solutions. In doing so, we aim to

- deepen the field's understanding of stakeholder tensions in spatial computing
- illustrates how technological and conceptual tools—like Apple Vision Pro hardware, SwiftUI coding, and design thinking exercises—can converge in workshop settings
- propose scalable strategies to harmonize these tensions in future deployments.

2 Background and Related Work

2.1 Spatial Computing as an Emerging Frontier

Spatial computing encompasses a range of technologies that integrate digital content with physical environments, including AR, VR, and MR platforms [4, 22]. Early conceptual work on AR highlighted the potential for blending virtual overlays with real-world settings [17]. Since then, advancements in hardware (e.g., head-mounted displays, depth-sensing cameras) and software (e.g., computer vision, real-time rendering) have accelerated the adoption of spatial computing in contexts such as education, simulation-based

training, and industrial maintenance. These technologies have enabled immersive and interactive experiences, significantly enhancing the applicability of spatial computing in diverse domains [12]. At the same time, researchers have identified emergent challenges related to usability, user experience, and ethical considerations, emphasizing that effective deployment requires more than technological sophistication alone [21, 22]. Specifically, ethical concerns have been raised about ensuring inclusivity and mitigating biases in spatial computing applications, which are critical for their responsible development [14]. Additionally, advancements in human-computer interaction (HCI) and real-time rendering technologies have laid the groundwork for immersive applications in industrial and educational contexts while spotlighting the importance of user-centric and ethical design frameworks [21]. Furthermore, spatial computing continues to evolve with integrated applications such as augmented reality (AR) and extended reality (XR), pushing the boundaries of what these technologies can achieve while confronting challenges like usability and ethical deployment [5]. For example, in industrial maintenance and training, interactive visualization tools and depth-sensing technologies have improved the operator experience significantly [19].

2.2 Participatory Approaches in HCI

Participatory Design (PD) has deep roots in human-computer interaction and cooperative design traditions. It foregrounds users' and stakeholders' experiential knowledge, advocating direct involvement in co-creating requirements and solutions. In workplace studies, researchers highlight how PD methods uncover hidden expertise and tacit knowledge, generating more robust design outcomes [8]. Applied to spatial computing, PD can be mighty in revealing how immersive interfaces shape social practices, user expectations, and organizational norms [6]. By engaging multiple stakeholders through iterative activities—such as collaborative prototyping and scenario-building—researchers can surface often-overlooked concerns, from privacy and accessibility to social acceptance and trust [23]. Recent empirical work demonstrates that participatory methods can adapt to different contexts, such as healthcare simulations [11], cultural heritage [10], or educational interventions [13]. PD also systematically addresses power imbalances among diverse user groups. Prototyping techniques, such as embodied design improvisation, have converted tacit knowledge into explicit, actionable insights [20]. Furthermore, enabling spaces for collaborative knowledge creation facilitates innovation and knowledge sharing [16]. These findings reinforce the role of PD in shaping inclusive and responsive design processes.

2.3 Positioning Our Work

Despite the growing body of literature on immersive technologies, few studies systematically examine how to identify and negotiate stakeholder tensions within spatial computing environments. Existing research has documented specific concerns (e.g., privacy or technical performance), yet comprehensive frameworks accounting for the interplay of divergent priorities across stakeholders remain scarce. For instance, participatory design has been proposed as a useful approach for engaging diverse stakeholders, revealing tensions between conflicting priorities such as user engagement and

technological performance[7]. Our work bridges this gap by using participatory methods to elicit and analyze the multifaceted challenges of designing for real-world spatial computing applications. Participatory mapping techniques, for example, have proven valuable in identifying spatial values and stakeholder preferences, though they often reveal underlying tensions and conflicts[3]. Additionally, “ecologies of contestation” highlight the agonistic spaces inherent in participatory design processes, emphasizing the need to balance competing interests[18]. Through workshops involving over 80 diverse participants, we reveal how tensions around digital-physical boundaries, privacy, accessibility, immersion, and innovation vs. user-centricity manifest and compete. These challenges align with prior research that situates participatory design as a process of navigating power dynamics and decision-making among stakeholders[2]. We thus expand upon earlier PD research by applying these methodologies to a domain where digital interactions are increasingly interwoven with everyday physical and organizational contexts[1]. By situating our study at this intersection, we aim to inform how stakeholders conceptualize spatial computing and how researchers and practitioners can collaboratively navigate conflicting incentives, ultimately guiding the design of more inclusive and harmonious immersive experiences.

3 Methodology

3.1 Study Design

We conducted eight online and two offline participatory workshops to explore how diverse stakeholders perceive and negotiate tensions in mixed reality environments. While we leveraged Apple Vision Pro devices and SwiftUI to prototype spatial interfaces, we emphasize that the design activities and analytic framework can be adapted to other platforms. By integrating digital and physical sessions, we aimed to capture a wide range of participant experiences and ensure inclusivity for those unable to attend in person.

3.2 Participant Recruitment

This study involved a diverse cohort of 83 participants over five workshops representing various stakeholders within the spatial computing ecosystem. The participant pool was strategically expanded to include many designers and design students, enhancing the representation of creative perspectives in the research. The participants were categorized into three main groups. Staff and End Users (n=34, 41%) comprised professionals who interact with spatial computing solutions in operational or daily contexts, including educators, administrative personnel, industrial workers, and other end-user professionals. Designers and Creative Practitioners (n=34, 41%), included UX/UI designers, industrial designers, graphic designers, design students at both undergraduate and graduate levels, and visual artists exploring digital mediums. The third group, Technologists and Engineers (n=15, 18%), consisted of software developers, hardware engineers, and systems integrators specializing in AR/VR/MR platforms.

Our main recruitment channels were professional networks, institutional mailing lists, and social media. Although snowball sampling allowed us to capture a rich diversity of participants,

we acknowledge potential biases, such as over-representing well-connected or tech-savvy individuals. Future work could employ more structured sampling to mitigate this limitation.

3.3 Hybrid Workshop Activities

This paper presents a novel hybrid workshop structure designed to introduce participants to cutting-edge spatial computing technologies, specifically focusing on Apple’s Vision Pro hardware and SwiftUI programming language. The workshop combines synchronous online sessions with intensive in-person experiences, offering a comprehensive learning environment that balances theoretical knowledge with hands-on practice. The workshop is structured as an 8-week program, incorporating both online and offline components:

3.3.1 Online Component.

- Duration: 60-100 minutes per session
- Frequency: Weekly (every Saturday)
- Platform: Utilizing video conferencing tools such as Zoom and Tencent Meetings
- Focus: Theoretical foundations including the history of spatial computing hardware and software, the technology behind the scenes, and scenario-based co-design workshop on future design goals of participants

3.3.2 Offline Component.

- Duration: Two full days (10:00 AM to 6:00 PM)
- Timing: Conducted over a weekend at the conclusion of the online sessions
- Location: In-person attendance required
- Day One Morning: Hands-on experience with Vision Pro hardware
- Day One Afternoon: Reflective sessions and analysis of the morning’s practical work
- Day Two Morning: Continued reflection and synthesis of learnings
- Day Two Afternoon: Coding camp for selected participants, focusing on SwiftUI for spatial computing applications

The workshop is designed to provide approximately 26 hours of total engagement per participant, distributed across the online and offline components. This intensive format allows for a deep dive into spatial computing concepts and technologies. The workshop also employs a blended learning approach, leveraging the strengths of both online and in-person instruction. The online sessions provide a foundation of knowledge and prepare participants for the hands-on components. The offline workshop intensifies the learning experience through direct interaction with advanced hardware and collaborative coding exercises.

3.3.3 Technological Focus. Two key technologies form the core of the workshop’s curriculum:

- Apple Vision Pro: Participants gain firsthand experience with this state-of-the-art spatial computing device, exploring its capabilities and potential applications.
- SwiftUI: Selected participants engage in a coding camp focused on this modern declarative UI framework, specifically in spatial computing development.

This hybrid workshop structure represents an innovative approach to technology education, particularly in the rapidly evolving field of spatial computing. By combining online theoretical instruction with intensive hands-on practice, the workshop aims to provide participants with a comprehensive understanding of both the hardware and software aspects of spatial computing. This model may serve as a template for future technology workshops, especially those dealing with emerging and complex technological ecosystems.

3.4 Data Collection & Analysis

We transcribed workshop discussions, coded them using a qualitative content analysis approach, and triangulated results with notes from prototyping outputs. Each transcript was independently reviewed by two researchers who assigned codes (e.g., “privacy concerns,” “access barriers,” “immersive overload”). Discrepancies were resolved in group meetings, resulting in a consolidated codebook. We then grouped codes into five overarching tensions described in our findings. Recognizing that these tensions often intersected (e.g., privacy concerns overlapping with accessibility challenges), we used iterative mapping exercises to illustrate these overlaps.

3.5 Ethical Considerations

Participants provided informed consent prior to workshops, and all identifying information (e.g., video, transcripts, design artifacts) was anonymized before analysis. Data was encrypted and stored on secure university servers, and only the core research team accessed the raw files.

4 Findings

Our analysis uncovered five critical tensions:

- (1) digital-physical boundaries
- (2) privacy and data ownership
- (3) accessibility and inclusivity
- (4) balancing immersion with situational awareness
- (5) reconciling technological advancement with human-centered design.

4.1 Divergent Perspectives on Digital-Physical Boundaries

“When the environment blurs too much, I don’t know what’s real or what I’m supposed to interact with. It can feel overwhelming.” (Staff participant, Workshop 3)

Staff and end users frequently called for clear demarcations between digital overlays and physical space, citing potential confusion or emotional discomfort. In contrast, some designers and technologists sought seamless integration, envisioning always-on overlays for continuous guidance or data-driven metrics. Privacy concerns sometimes emerged here, with participants questioning whether always-on cameras or sensors threatened personal boundaries.

4.2 Privacy and Data Ownership Concerns

“If these headsets function using VST (video see-thru) technology, as you mentioned in the online workshop, I’m worried these headsets can record everything—audio, video, my location. Who owns that data?” (End User participant, Workshop 4)

Institutional staff voiced apprehension about compliance with regulatory policies, while end users were uneasy about the potential misuse of personal images or location data. Conversely, some technologists initially downplayed these issues, prioritizing functional performance. After iterative discussions, however, they acknowledged the long-term consequences of large-scale data collection and championed more explicit usage guidelines (e.g., data storage transparency and user consent protocols).

4.3 Accessibility and Inclusivity of Spatial Interfaces

“We can’t assume everyone has the latest hardware or perfect vision. If we don’t plan for that, we leave people behind.” (Designer participant, Workshop 2)

Participants highlighted socioeconomic barriers (e.g., high headset costs) and physical constraints (e.g., low-vision users). Designers proposed layered accessibility, offering both advanced immersions for those with high-end hardware and simplified 2D versions for essential devices. When participants explored Apple Vision Pro prototypes, some found them too expensive or cumbersome for large-scale institutional adoption; this prompted discussions about how to adapt these solutions for more affordable platforms.

4.4 Balancing Immersion with Situational Awareness

“If I’m fully immersed, I could miss what’s going on around me—like students needing help.” (Staff participant, Workshop 3)

Technologists and designers viewed deep immersion as a key differentiator of spatial computing. Yet staff in educational and industrial settings worried about safety hazards and real-time situational needs (e.g., workers noticing conveyor belt issues, teachers monitoring students). Proposed solutions included “off-switches” for fast transitions between immersive and semi-immersive modes and integrated warning overlays for essential real-world alerts.

4.5 Trade-off Between Technological Advancement and Human-Centered Design

“We can keep pushing hardware specs, but if it makes people feeling too heavy, nauseous or anxious, that’s a fail.” (Technologist participant, Workshop 1)

Some technologists championed rapid iteration on sensor fusion, machine learning, and rendering fidelity, while others, including designers and staff, stressed usability, safety, and emotional well-being. This tension highlighted a common dilemma in emerging tech: pushing innovation without overlooking fundamental human-centric needs. Many participants voiced concerns about the over-weight of the device and “feature creep,” urging careful cost-benefit evaluations of each new capability.

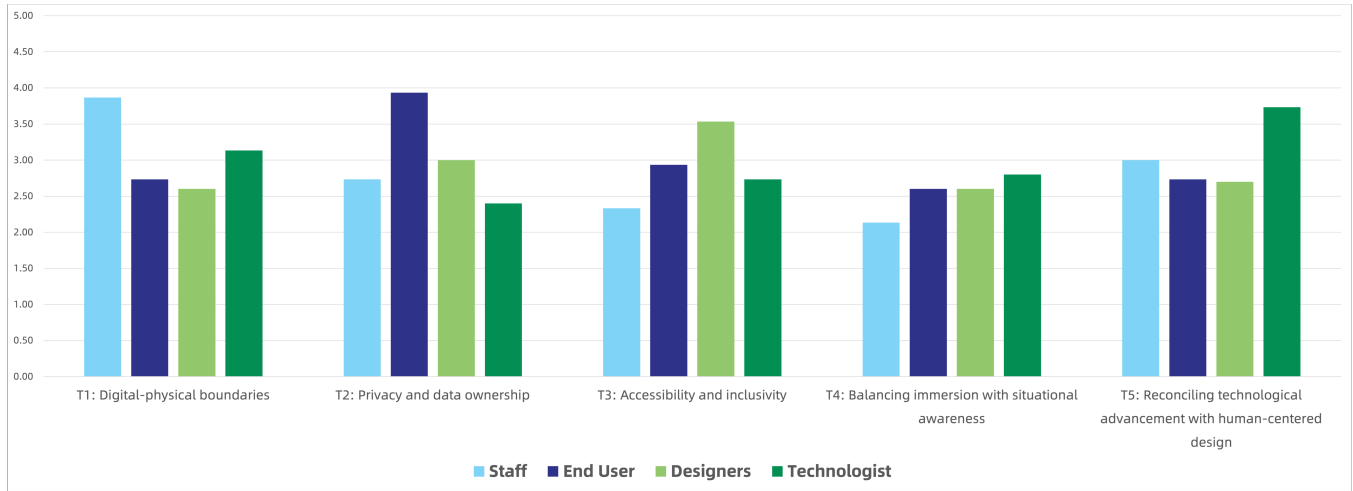


Figure 2: Perceived importance of tensions in technology design across stakeholder groups. The five tensions include T1: Digital-physical boundaries, T2: Privacy and data ownership, T3: Accessibility and inclusivity, T4: Balancing immersion with situational awareness, and T5: Reconciling technological advancement with human-centered design. Technologists consistently rate these tensions higher compared to other groups, particularly for T5 and T3.

5 Discussion

5.1 Intersections and Complexities of Stakeholder Tensions

The five tensions rarely manifested in isolation. Privacy (T2) is interlinked with digital-physical boundaries (T1), as always-on sensors capture data that may compromise user comfort in merged environments. Similarly, accessibility (T3) and immersion (T4) overlapped when participants discussed ensuring that immersive features do not hinder those with limited hardware access or unique physical constraints. Recognizing these intersections is crucial for more holistic solutions. Understanding the diverse perspectives of stakeholders is essential in addressing key tensions in technology design. Figure 2 illustrates how four stakeholder groups—staff, end users, designers, and technologists—perceive the importance of five critical tensions in design processes. These visualizations underscore the complexities in aligning stakeholder priorities. While technologists prioritize forward-looking aspects like accessibility (T3) and advancing technology with human-centered approaches (T5), end users focus less on these issues. Such differences necessitate collaborative strategies to balance competing priorities effectively.

5.2 Feasibility of Contextual Customization and Layered Accessibility

Our findings suggest contextual customization—allowing users to tailor the level of immersion or data capture to specific tasks or environments—can mitigate many tensions (e.g., toggling overlays in high-risk industrial settings). Similarly, layered accessibility (e.g., multi-modal feedback, simplified 2D interfaces) ensures broader inclusivity. However, implementing these approaches requires both technical infrastructure (flexible software frameworks) and organizational commitment (budget, time for user training).

5.3 Organizational and Power Dynamics

Designers and staff noted concerns about power imbalances—for example, technology vendors dictating data policies without fully engaging institutional voices. Participatory design methods can shift this dynamic by giving stakeholders direct input on critical decisions, such as privacy settings or user onboarding sequences. Nevertheless, company hierarchies and contractual obligations can still impede balanced collaboration. Future work might explore embedding stakeholder alignment strategies in procurement processes or governance frameworks.

5.4 Implications for Spatial Computing Design

The tensions uncovered in our study underscore the need for holistic design frameworks that integrate stakeholder perspectives early and iteratively. Traditional user-centered design approaches may partially address these needs. Still, the rapid evolution of spatial computing, coupled with ethical and sociotechnical complexities, demands that designers, technologists, and organizations collaborate in ongoing dialogue:

- **Contextual Customization:** Rather than enforcing a universal “optimal” level of immersion or technological sophistication, designers might provide adjustable settings that let users tailor immersion levels or privacy configurations to specific tasks and contexts.
- **Transparent Data Practices:** Organizations adopting spatial computing should communicate data practices (e.g., usage, storage, and ownership) in plain language and develop co-created guidelines with technical teams and end users. This clarity can mitigate anxieties around privacy (T2) and reinforce trust.
- **Layered Accessibility:** Offering “accessibility layers” or adaptive interfaces can help accommodate a wide range of user

needs and device capabilities (T3). For instance, overlay-based AR systems might include simplified 2D “mirroring” for users with limited visual immersion hardware.

- **Incremental Adoption Strategies:** Rather than immediately deploying advanced MR systems enterprise-wide, phased rollouts with continuous user feedback enable organizations to address concerns around safety, user comfort, and governance progressively.

6 Conclusion and Future Directions

In this Late-Breaking Work, we examined five tensions when designing mixed reality systems involving diverse stakeholders. By integrating technological explorations (Apple Vision Pro, SwiftUI) with conceptual frameworks (participatory design, scenario exploration, co-creation), we uncovered the nuanced ways privacy, accessibility, immersion, and innovation trade-offs intersect.

Future research could apply our participatory methods to additional sectors, such as healthcare and public governance, or conduct cross-cultural studies to see how local regulations and social norms reshape these tensions. Longitudinal studies would also help track how tensions evolve as organizations move from prototypes to widespread implementation—illuminating strategies to sustain user trust, data ethics, and inclusive practices.

Ultimately, our findings underscore the need for collaborative, ongoing engagements that reflect stakeholder voices throughout the design lifecycle. By balancing rapid technical innovation with ethical and human-centered considerations, spatial computing solutions can achieve broader acceptance and positive social impact across industries.

Our study’s scope was constrained by workshop duration and participant recruitment methods, potentially overlooking perspectives from additional domains (e.g., healthcare, public governance) or underrepresented user groups with specific access needs. Future research could explore longitudinal studies to observe how tensions evolve, particularly as organizations integrate immersive technologies into routine operations. Furthermore, cross-cultural investigations might reveal distinct societal or regulatory nuances, offering broader insight into how stakeholder tensions manifest in varied global contexts. Finally, more profound exploration into toolkits and guidelines for participatory design in spatial computing could empower practitioners to address stakeholder tensions from the earliest stages of conceptualization systematically.

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